Earthquakes



Information

Earthquake= a sudden and rapid movement of the ground.

Measuring

Earthquakes are measured by machines called Seismometers. The unit of measure is points on the Richter scale. Each time the Richter scale increases by one the force of the earthquake increases by ten.

Waves

The shaking occurs in waves. These waves are called Seismic waves. There are three types of Seismic wave:
P-waves=these waves move backwards and forwards (longitudinally) in the Earth.
S-waves=these waves move up and down (transversely) in the Earth.
Surface waves=waves moving along the Earth's surface. These cause the most damage.

Where they start

Earthquakes occur most often at fault lines.
Focus=the place and earthquake starts under the ground.
Epicentre=the place on the surface of the Earth directly above the focus. This is where most of the energy is released.

Effects

- The ground surface deforms (changes shape).
- Buildings fall down. Most earthquake-related deaths are caused by the collapse of structures.
- Bridges fall down.
- Landslides where large pieces of land slide downwards
- Tsunamis, which is when large waves are caused by the earth moving under water
- Fires started by gas and other fuel lines being broken.
- Sewage pipes burst. Apart from being smelly this increases the chances of people getting sick after the earthquake.
- Water pipes for drinking water burst.
- Flooding caused as dams burst.
- Breakdown in communication systems, both because wires are damaged and there is no power.
- Transport networks disrupted.
- Power supplies disrupted.
- Looting.

Questions

Earthquake= a sudden and r_____ movement of the ground.

Measuring

Earthquakes are measured by machines called S_____. The unit of measure is points on the R_____ scale. Each time the Richter scale increases by one the force of the earthquake increases by t_____.

Waves

The shaking occurs in _____. These waves are called S_____ waves. There are three types of Seismic wave:

P-waves=these waves move backwards and f_____ (longitudinally) in the Earth. Draw a diagram of a P wave:

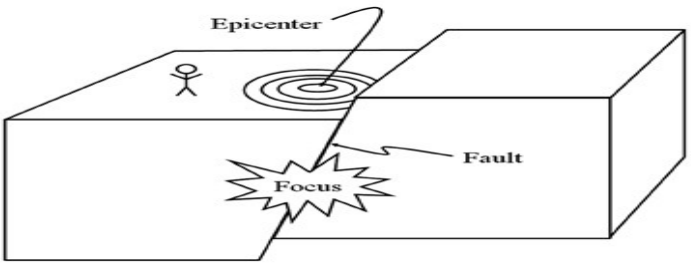
S-waves=these waves move up and down (t_____) in the Earth. Draw a diagram of an S wave:

Surface waves=waves moving along the Earth's _____. These cause the most damage. Draw a diagram of a surface wave:

Where they start

Earthquakes occur most often at _____ lines.

Focus=the place and earthquake _____ under the ground.



Epicentre=the place on the surface of the Earth directly above the _____. This is where most of the energy is released.

Why is the epicentre not always the place where the fault can be seen on the surface?

Effects

- The ground surface d_____ (changes shape).
- Buildings fall down. Most earthquake-related deaths are caused by the collapse of _____.
- Bridges fall down.



- L_____
- Large waves called _____ may hit coastal areas
- Fires started by _____ and other fuel lines being broken.
- Sewage pipes burst. Apart from being smelly this increases the chances of people getting _____ after the earthquake.
- _____ for drinking water burst.
- Flooding caused as _____ burst.
- Breakdown in _____ systems.
- As roads are damaged or have rubble on them _____ are disrupted
- Power supplies disrupted
- Looting.





Why do you think power supplies are disrupted?

What may be some effects of burst water pipes?

What do you think looting means?

Why do you think looting happens at this time?

Irpinia earthquake

At the epicentre of and earthquake the release of energy on the surface is greatest. As the energy moves away from the epicentre the waves spread out in concentric circles. As the waves move they lose energy. Look at the diagram below and use it to answer the questions that follow.

The 1980 Irpinia earthquake took place in the Irpinia region in Southern Italy on Sunday, November 23, 1980. Measuring 6.89 on the Richter Scale, the quake, centered on the village of Conza, killed 2,914 people, injured more than 10,000 and left 300,000 homeless.



Place the following towns and cities in order of which would have received the greatest seismic force from the Earthquake. Give a reason for your answer.
Napoli(Naples), Rome, Potenza, Conza and Siena.

Why did more people die in the towns than in the country? Give two reasons.

Earthquake

D F R R E T H C I R R Y T F S
Q I P WL E O A G E D L S A M
G E L S G L B F T J G E U U A
Z H WA L R P E L A C S N L D
Z E WA F O MN M V C R A T R
Y E P O V O L WR S Z E MI O
S S C L ME WX O J B V I Y Q
E UWS A E S X F G V S S WG
S Z E K X NC Y E L S NJ Y Q
T I UNO S D A D Y B A F P G
S OI XB HF S F L Z R D A H
A R O Q A K D A L R B T X S X
N K WL U HH G B I UL F MM
E P I C E NT R E Y D S K U Y
N R A P I DX B K S Y E C Y K

COLLAPSE
DAMS
DEFORM
EPICENTRE
FAULT
FOCUS
LANDSLIDE
LOOT
RAPID
RICHTER
SCALE
SEWAGE
SIESMOMETER
SURFACE
TRANSVERSELY
TSUNAMIS
WAVES

